

Mark as done



Now you have submitted your answers, compare your own answers to those below. Remember - there are different ways to answer these questions, so your answers may look different to those below, which are just examples of one possible approach. Select **Answer** to see the model answers.

Answer

Now review the model answer below and note how your solution compares.

1. You should see that if $|A| = n$, then $|\wp(A)| = 2^n$, i.e. there are 2^n subsets of A . To prove this we first enumerate the elements of A , $A = \{a_1, a_2, \dots, a_n\}$. For an arbitrary subset $B \subseteq A$, for each element a_i either $a_i \in B$, or $a_i \notin B$. Since a set is determined by its elements, there are n binary options for the elements of B , giving us $2 \times 2 \times \dots \times 2 = 2^n$ different options for B . Therefore there are 2^n different subsets of A , giving $|\wp(A)| = 2^n$.

2.

$$\begin{aligned}
 x \in A \cap (B \cup C) &\leftrightarrow x \in A \wedge x \in (B \cup C) \\
 &\leftrightarrow x \in A \wedge (x \in B \vee x \in C) \\
 &\leftrightarrow (x \in A \wedge x \in B) \vee (x \in A \wedge x \in C) \text{ (due to the definitions of } \wedge \text{ and } \vee) \\
 &\leftrightarrow (x \in A \cap B) \vee (x \in A \cap C) \\
 &\leftrightarrow x \in ((A \cap B) \cup (A \cap C))
 \end{aligned}$$

3. Note that $A \cup B$ can be split into three sets $A \setminus B$, $B \setminus A$ and $A \cap B$, which each have no overlap with each other. Therefore we have $|A \cup B| = |A \setminus B| + |B \setminus A| + |A \cap B|$. Likewise, we can split A into the non-overlapping sets $A \setminus B$ and $A \cap B$, and B into $B \setminus A$ and $A \cap B$, giving us $|A| = |A \setminus B| + |A \cap B|$ and $|B| = |B \setminus A| + |A \cap B|$. Therefore we have:

$$\begin{aligned}
 |A \cup B| &= |A \setminus B| + |B \setminus A| + |A \cap B| \\
 &= (|A| - |A \cap B|) + (|B| - |A \cap B|) + |A \cap B| \\
 &= |A| + |B| - |A \cap B|
 \end{aligned}$$

